

Practical – Week2

1. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

```
pawan_kumar@LAPTOP-SSD0PUQ2:~$ nano filecopy.c
pawan_kumar@LAPTOP-SSD0PUQ2:~$ echo "This is the content of the source file." > source.txt
pawan_kumar@LAPTOP-SSD0PUQ2:~$ cat source.txt
This is the content of the source file.
pawan_kumar@LAPTOP-SSD0PUQ2:~$ gcc filecopy.c -o filecopy
pawan_kumar@LAPTOP-SSD0PUQ2:~$ ./filecopy
File copied successfully.
pawan_kumar@LAPTOP-SSD0PUQ2:~$ cat destination.txt
This is the content of the source file.
pawan_kumar@LAPTOP-SSD0PUQ2:~$ ls -l source.txt destination.txt
-rw-r--r-- 1 pawan_kumar pawan_kumar 40 Aug 17 05:05 destination.txt
-rw-r--r-- 1 pawan_kumar pawan_kumar 40 Aug 17 05:04 source.txt
pawan_kumar@LAPTOP-SSD0PUQ2:~$ diff source.txt destination.txt
pawan_kumar@LAPTOP-SSD0PUQ2:~$ |
```

Report: I compiled and ran `filecopy.c` to copy `source.txt` to `destination.txt`. The program reported "File copied successfully," and `cat destination.txt` showed the content matched the source exactly. `ls -l` confirmed both files were the same size (40 bytes), and `diff source.txt destination.txt` produced no output, indicating no differences. This confirms the program correctly reads from the source file and writes an identical copy to the destination file.

2. Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved


```

pawan_kumar@LAPTOP-SSDC x + v
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7bd95fe45cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "55cdc4cc000-55cdc45be000 r--p 0", 1024) = 1024
read(3, "fdfe000-7bd95fdfff000 r--p 000a90", 1024) = 1024
read(3, "00 00:30 13650", 1024) = 1024
read(3, " /usr/lib/x86_64-linux", 1024) = 1024
read(3, "0 13644 /us", 1024) = 566
close(3) = 0
sched_getaffinity(1120, 32, [0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15]) = 8
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7bd9601cf000
mprotect(0x7bd9601cf000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x7bd9601d0000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid(0) = 1120
rt_sigaction(SIGSEGV, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, NULL, 0) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, NULL, 0) = 0
prctl(PR_GET_AUXV, 0x7ffcc740d38, 0x200, 0, 0) = 448
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/uucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "# Common strings shared across a", 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7bd95fe45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, 0) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN_RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7bd95fe45cb0}, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN_RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7bd95fe45cb0}, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, 0) = 0
ioctl(3, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|AB0|BS0|VT0|FF0|POST|ONLCR, c_cflag=B38400|CS0|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHONL|TEXTEN|ECHOCTL|ECHOK, ...}) = 0
brk(0x55cdc5f5e000) = 0x55cdc5f5e000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=25, ...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=25, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(0, F_SETPIPE_SZ, 1048576) = 1048576
splice(0, NULL, 5, NULL, 1048576, 0) = 25
splice(4, NULL, 1, NULL, 25, 0Hello Linux System Calls) = 25
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x7bd9601cf000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
pawan_kumar@LAPTOP-SSDCPUQ2:~$

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pawan_kumar@LAPTOP-SSDC x + v
rt_sigaction(SIGBUS, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, NULL, 0) = 0
prctl(PR_GET_AUXV, 0x7ffcc740d38, 0x200, 0, 0) = 448
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/uucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "# Common strings shared across a", 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7bd95fe45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, 0) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN_RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7bd95fe45cb0}, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN_RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7bd95fe45cb0}, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, 0) = 0
ioctl(3, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|AB0|BS0|VT0|FF0|POST|ONLCR, c_cflag=B38400|CS0|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHONL|TEXTEN|ECHOCTL|ECHOK, ...}) = 0
brk(0x55cdc5f5e000) = 0x55cdc5f5e000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=25, ...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=25, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(0, F_SETPIPE_SZ, 1048576) = 1048576
splice(0, NULL, 5, NULL, 1048576, 0) = 25
splice(4, NULL, 1, NULL, 25, 0Hello Linux System Calls) = 25
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x7bd9601cf000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
pawan_kumar@LAPTOP-SSDCPUQ2:~$

```

Report: I ran strace cat sample.txt. The trace showed openat() to open the file, fstat() for metadata, splice() to transfer data directly to stdout, and close() to release the descriptor. This confirms cat uses efficient kernel calls like splice() rather than a plain read/write loop.